

Module 11: Error Correction & Hamming Code Architecture

Unit 2: Data Link Layer & MAC Sublayer

Forward Error Correction (FEC)

Mathematical principles of error correction: Hamming distance theorems, redundancy inequality $2^r \geq m + r + 1$, parity placement at powers of 2, and syndrome word error location decoding.

1. Bit Positions & Parity Check Sets (7-Bit Codeword)

Pos 1

Pos 2

Pos 3

Pos 4

Pos 5

Pos 6

Pos 7

p1

p2

d1

p4

d2

d3

d4

001

010

011

100

101

110

111

• Parity Bit Mathematical Coverage (Even Parity):

p1 (Bit 0 in pos is 1) → Checks: 1, 3, 5, 7 → $p1 = d1 \oplus d2 \oplus d4$

p2 (Bit 1 in pos is 1) → Checks: 2, 3, 6, 7 → $p2 = d1 \oplus d3 \oplus d4$

p4 (Bit 2 in pos is 1) → Checks: 4, 5, 6, 7 → $p4 = d2 \oplus d3 \oplus d4$

Parity bits placed strictly at powers of 2 (1, 2, 4, 8, 16...)

2. Syndrome Decoding Word (S4 S2 S1)

Syndrome Bit Equations at Receiver:

$s1 = b1 \oplus b3 \oplus b5 \oplus b7$

$s2 = b2 \oplus b3 \oplus b6 \oplus b7$ | $s4 = b4 \oplus b5 \oplus b6 \oplus b7$

Syndrome Interpretation & Correction:

If Syndrome = 000 → No error! Data accepted.

If Syndrome ≠ 000 → Binary value equals the ERROR BIT position!

E.g., Syndrome = 110 (binary 6) → Bit 6 is corrupted!

Correction Action: Simply invert bit 6 (0 → 1 or 1 → 0)!

No retransmission required! Forward Error Correction (FEC).

Hamming Distance Theorem & Redundancy Inequality (AKTU Core Formulas)

Redundancy Bits Formula:

$2^r \geq m + r + 1$

where m = data bits, r = check/parity bits. For m = 4 bits: $2^3 = 8 \geq 4 + 3 + 1 = 8$ (r = 3).

Hamming Distance d_{min} :

Number of bit positions in which two codewords differ (XOR weight).

Error Detection Condition:

$d_{min} \geq d + 1$ (To detect up to 'd' bit errors)

Error Correction Condition:

$d_{min} \geq 2t + 1$ (To correct up to 't' bit errors)

1. Hamming Distance & FEC

Forward Error Correction allows the receiving terminal to detect and automatically correct bit flips without requiring retransmission. A code with minimum distance d_{min} can detect d errors if $d_{min} \geq d + 1$ and correct t errors if $d_{min} \geq 2t + 1$.

2. Hamming (7, 4) Syndrome Decoding

Syndrome Word Error Locator

$$S1 = b1 \oplus b3 \oplus b5 \oplus b7$$

$$S2 = b2 \oplus b3 \oplus b6 \oplus b7$$

$$S4 = b4 \oplus b5 \oplus b6 \oplus b7$$

$$\text{Syndrome } (S4 \ S2 \ S1)_2 = \text{Error Bit Index (1 to 7)}$$